

September 6, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript, "Physics-Exact Credit Assignment for Multi-Agent Flutter Suppression," for consideration as a Regular Article in AIAA Journal.

This manuscript was previously submitted to the Journal of Guidance, Control, and Dynamics, whose Editor-in-Chief withdrew it as outside that journal's scope (a distributed-control contribution validated on an aeroelastic testbed, rather than a guidance, navigation, or flight-control result in the narrower sense JGCD covers). We believe AIAA Journal, whose scope spans aerospace structures, dynamics, and control without that restriction, is the right venue for a paper that sits deliberately across those areas.

The paper studies active flutter suppression on a wing with several independently actuated trailing-edge surfaces, each commanded by its own reinforcement-learning agent from local sensing alone – the setting a physically distributed avionics architecture actually presents, as opposed to the single-agent, single-effector, globally-observed setting the existing flutter-suppression literature uses throughout. Posing the problem this way exposes a credit-assignment question with a physical answer: for this class of energy-conserving structural plant, we show that the per-agent credit multi-agent reinforcement learning normally has to estimate is instead available in closed form directly from the equations of motion, and we validate it against a naive undecomposed reward and an adapted value-decomposition baseline (Holm-Bonferroni corrected across the full comparison family), against a six-rung classical baseline ladder (including a fully decentralised LQG synthesised from the same sensing the learned agents have), and against an unsteady vortex-lattice solver for the underlying plant. A second, more general finding follows from the same study: on this problem, the initial policy-exploration scale dominated every algorithmic choice we tried, by a factor of two, and is computable in advance from a classical design – a diagnostic we believe generalises to other physical control tasks with a narrow useful actuation band. We report two negative results (learned communication did not help; a shaping-based refinement did not survive correction for multiple comparisons once compared against the full family of alternatives) at the same statistical standard as the positive ones.

We believe this manuscript is a strong fit for AIAA Journal: it derives a result from the structural-dynamics equations of motion of an aeroelastic plant (a Goland wing), validates it against an unsteady vortex-lattice solver and a classical decentralised control baseline, and uses that physically grounded testbed to answer a distributed-control and credit-assignment question that a single-discipline aerodynamics, structures, or guidance-and-control journal is not positioned to referee on its own. It also engages directly with the certification gap between a learned policy and a classical design with a known stability margin, which we discuss explicitly in the paper.

Other than the JGCD submission disclosed above, this manuscript, in whole or in substantial part, has not previously been submitted to any AIAA journal or conference. I will provide any further disclosure ScholarOne requests. I declare no conflicts of interest, and no funding was received for this work. Data and code needed to reproduce every result are cited in the manuscript's Data and Code Availability section.

Thank you for your consideration. I look forward to your response.

Sincerely,

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